Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Phase 1: Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named "SDC2026_TeamName_Preliminary.pdf" before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

1. Team & Project Overview

Team Name	NoBugPls
Selected Domain	Domain 1 - The Aiversecity Future-Ready Workforce Domain
Selected SDG Track (1–17)	SDG 8: Decent Work and Economic Growth (primary); SDG 4: Quality Education (supporting)
Project / Solution Name	AiversePulse Enterprise
Name (Leader)	TANG YU SHAN
Email & Phone (Leader)	tangys2005@gmail.com & +60175908412
University Affiliation(s)	Universiti Teknologi Malaysia (UTM); Universiti Tunku Abdul Rahman (UTAR)
Number of Team Members	5

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Employers expect 39% of workers' core skills to change by 2030 (WEF Future of Jobs Report 2025). Employees whose roles are being reshaped by AI, and the HR teams who must decide where to spend limited training budgets, have no transparent way to see which skills are rising, which are falling, and which future roles a person can realistically reach. Today's tools are static job descriptions, annual skills surveys, and generic course catalogues. None of them show the evidence behind a recommendation.

Individual learners face the same gap. Mid-career staff and career changers cannot easily judge whether a move is realistic, which skill gaps matter most, or how many hours a week they must commit. Poorly targeted reskilling wastes time and money and can leave people in roles at risk of automation.

The people affected are workers who need a decent job after their role changes. This links the problem to SDG 8 (Target 8.5: full and productive employment and decent work for all) and SDG 4 (Target 4.4: relevant skills for employment and decent jobs). HR advisors need the same answers for their teams, and they must be able to justify every recommendation to the people it affects.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

AiversePulse is a multi-agent reskilling assistant with two views: a public app for individual learners and an HR console for workforce planners. A LangGraph supervisor routes each request. Profiler, knowledge-retrieval (FastMCP) and market-analyst agents run in parallel. A Gap Analyst compares the person's skills with each target role under two AI scenarios: Assistive AI (requirement levels as they stand) and Agentic AI (judgement and oversight weigh more). A Pathway agent builds a 30/60/90-day plan capped to the learner's weekly hours. A Reviewer checks every claim against its cited evidence and allows at most two targeted repairs.

Role matching runs live across ~3,000 ESCO v1.2 occupations; a requirement model trained offline on O*NET 31.0 transfers skill importance and levels to ESCO skills. An STF-KAN forecaster, trained on O*NET release history 25.0 to 31.0, ranks rising skills and is served through an MCP tool. A daily AI-trend board built from public sources is dated and evidence-backed. SDG links use sentence embeddings against the 169 UN SDG targets and are shown only when the match is confident.

Plans stay drafts until a named human approves them, and each version is kept in an immutable audit log. The HR console adds screening agents driven by plain-language thresholds, blind review of redacted resumes, and a human shortlist decision. A RAG chat agent answers follow-up questions with cited passages; a Three.js 3D hologram interviewer runs adaptive mock interviews.

Workshop skills applied: GenAI APIs (LLM routing and drafted explanations, optional so the app runs without it), database architecture (SQLite audit trail, ChromaDB vector store) and cloud deployment (Replit frontend, dual-node VPS backend, standalone RunPod MiniLM inference, Docker).

Limitations stated in the app: forecasts are historical backtests, and the rising-skill signal is modest (AUC 0.568 on held-out data), so it ranks skills rather than predicting them.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Language and contracts: Python 3.11+, Pydantic v2

Agents: LangGraph StateGraph (supervisor, profiler, knowledge search, market analyst, gap analyst, pathway, coach, reviewer), plus resume-screening, RAG chat and mock-interview agents

Tools: FastMCP server; ChromaDB with MiniLM embeddings (all-MiniLM-L6-v2) for RAG over 18,444 ESCO and learning passages, powering the RAG chat agent

Forecasting: PyTorch STF-KAN, trained offline on O*NET release history

Data: ESCO v1.2 and UN SDG API (runtime); O*NET 31.0 (training only); HN, arXiv, Arbeitnow and Remotive daily trend feeds; SQLite for plans, approvals and audit

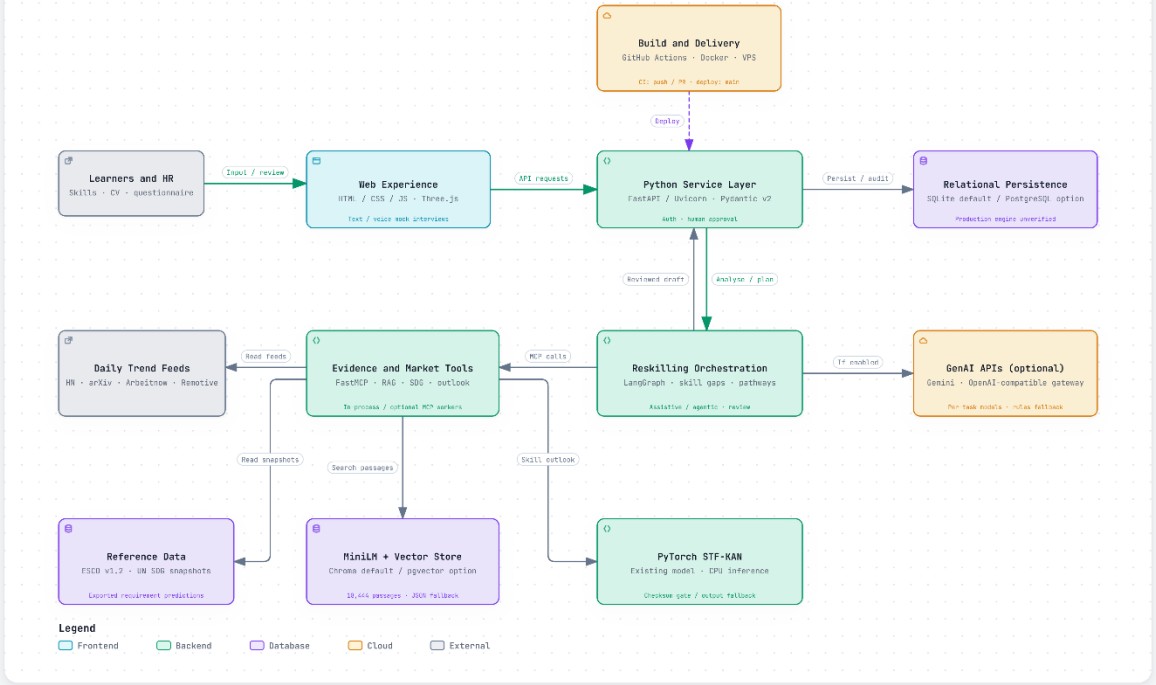
Web: FastAPI with plain HTML/CSS/JS (Streamlit kept as a fallback); Three.js 3D hologram interviewer with text or browser-voice answers

LLM (optional): Google Gemini 3.1 Flash-Lite via google-genai, plus an OpenAI-compatible gateway for per-task models (coach, resume screening, voice input); the app is deterministic without them

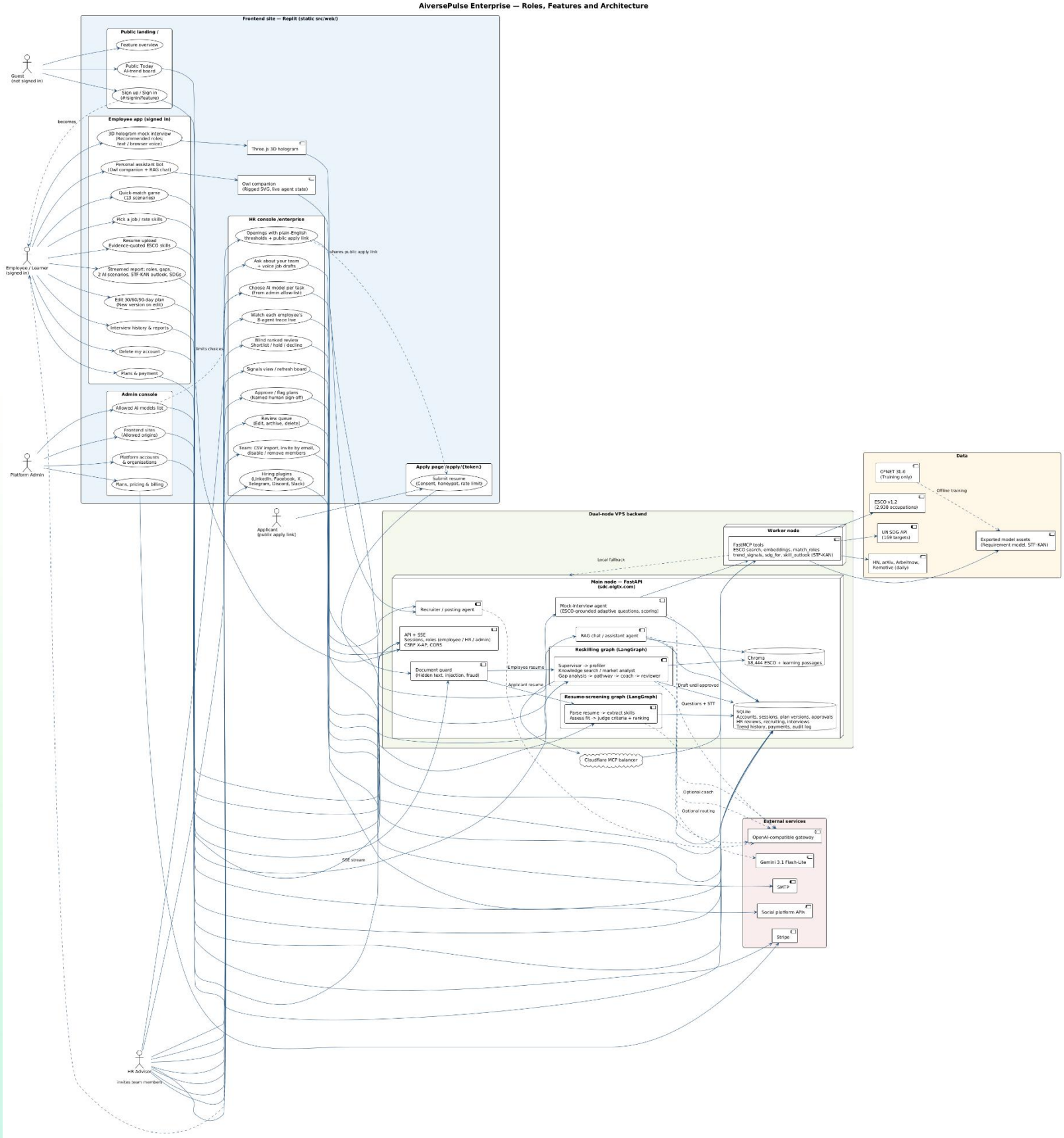
Deployment: backend API on a dual-node VPS (main server + worker node behind a Cloudflare MCP balancer); Replit (development and frontend site); RunPod NVIDIA A40 Pod for a separate pretrained MiniLM inference experiment, Dockerfile (container); pytest

Layer	Technology	Purpose
Core Backend	Python 3.11+, Pydantic v2	System logic and more structured data validation
AI Orchestration	LangGraph StateGraph	Coordination of profiler, knowledge, market, gap-analysis, pathway, coach and review agents, plus resume screening
Evidence Retrieval	FastMCP + Chroma DB	Retrieves attributed ESCO and workforce knowledge passages for the RAG chat agent and serves daily AI-trend evidence.
Semantic Matching	MiniLM embeddings (all-MiniLM-L6-v2)	Compares related skills, occupations and SDG content
Forecasting	PyTorch STF-KAN	Analysis of historical changes in skill demand
Occupational Data	O*NET 31.0	Skill and occupation requirements (ESCO v1.2 at runtime; O*NET 31.0 for training only)
Database	SQLite	Stores pathways, approvals and audit records
Web	FastAPI + HTML/CSS/JS	Runs the user and HR interfaces, including the Three.js 3D interview hologram
LLM	Gemini 3.1 Flash-Lite + OpenAI-compatible gateway	Optional routing, coaching explanations, resume skill proposals and mock interviews.
Deployment	Replit	Development environment and frontend site; the backend API runs on a dual-node VPS (main + worker node)
GPU Compute	RunPod	Standalone inference experiment using pretrained MiniLM to generate embeddings for six example career skill descriptions. No training, fine-tuning or website integration.
Testing	pytest	74 pytest tests plus Node UI tests, run by GitHub Actions CI on every push

AiversePulse Conceptual Technology Stack



AiversePulse Enterprise — Roles, Features and Architecture



Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Yes. Replit is used as the development environment and hosts the AiversePulse frontend at https://sdc-hackathon-submission-no-bug-pls.replit.app/, which calls our backend API on a dual-node VPS. Replit proof of use is attached to this proposal. RunPod was also used for a standalone GPU inference experiment on an NVIDIA A40 Pod. The pretrained sentence-transformers/all-MiniLM-L6-v2 model converted six example career skill descriptions into 384-dimensional embeddings. No STF-KAN or requirement-model training was performed on RunPod, and the experiment was not integrated into the deployed application. RunPod usage evidence will be included in the GitHub repository.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

We started from defined users, not a model. Our product brief names two users (learners and HR advisors) and five design principles, including "show the evidence, not the confidence" and "humans hold the pen". Three intake routes (a 13-scenario quiz, resume upload, or picking a target job) reflect that people know their skills to different degrees. Resume text is converted to sanitized Markdown, with hidden text and prompt-injection lines stripped. Personal details are redacted before HR review.

Technical viability was tested during the build. The app runs end-to-end without a GPU or API key, every agent has a deterministic fallback, and each must-have maps to a named test. Deployment readiness: a live dual-node VPS backend with a Replit-hosted frontend, a Dockerfile, seeded demo accounts, an offline demo mode and CI tests on every push.

Limitation: we have not yet run formal user testing with learners or HR advisors. Before Phase 2 we plan user testing, a production-grade deployment, and a pilot that measures plan completion.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	**Primary**: Target 8.5, achieve full and productive employment and decent work for all women and men; the app also uses it as its baseline when no confident SDG match exists. **Supporting**: Target 4.4, substantially increase youth and adults with relevant skills for employment, decent jobs and entrepreneurship.

Metric Category	Your Response
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Employers expect **39%** of workers' core skills to change by 2030, down from 44% in 2023.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	% of plan activities that cite a dated source and an SDG target, checked by the Reviewer. Secondary: rising-skill lift over base rate, currently 1.39x on held-out O*NET data (base rate 2.1%).
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	≥95% of plan activities pass the evidence check on first draft, and 100% of activities shown for approval are either verified or flagged. Keep rising-skill lift ≥1.39x after the O*NET data refresh.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

☑ All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.

☑ The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.

☑ The proposed solution targets a functional, deployed application rather than a theoretical business case study.

☑ All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☑ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☑ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
O*NET 31.0	dataset	CC BY 4.0 (USDOL/ETA)
sentence-transformers/all-MiniLM-L6-v2	model	Apache-2.0
FastMCP	code	Apache-2.0
pdf-prompt-injection-toolkit	code	MIT
ESCO v1.2 occupations and skills (European Commission)	dataset	EC reuse policy, attribution required
UN SDG goals and targets (UN Statistics SDG API)	dataset	© United Nations; UN terms of use
LangGraph;	code	MIT;
FastAPI;	code	MIT;
PyTorch;	code	BSD-3-Clause;
ChromaDB	code	Apache-2.0

Gemini 3.1 Flash-Lite; gateway models (grok-4.7, gpt-5.6-luna, mimo-v2.6-flash)	model (API)	Provider API terms
Three.js; GSAP / ScrollTrigger	code	MIT; GSAP Standard License

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

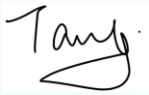
☐ Yes ☑ No

If yes, please describe the nature of the agreement and which member it concerns:

---

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team's knowledge and that the project will be original work created during the SDC 2026 hackathon period.



[Signature]

24/10/2026

[Date]